

TQEC 2/18

Need more people to serve as a backup screen recorder

Junyu: TopoLS presentation (w/ paper)

Transform quantum circuit into lattice surgery based operations

Optimization workflow maps quantum circuits to pipe diagrams

Embed ZX diagrams

Construct via smaller blacks, making it scalable

Perform optimizations via ZX representation

Layer slicing: index ZX nodes in the graph, create embedding order (-> MCTS representation)

Layer slices are mapped into circuit layout with simpler structure

MCTS for 3d embedding: create hierarchical tree structure to represent the embedding

Monte Carlo search to find the embedding

Expand diagrams into 3D space (time/space axes), make connections based on the ZX graph to show state evolution

Use shortest path function to find the connections

Color switching: modify inputs to reshape circuit components

Use topology aware circuit partitions to enable fixed space regions to align the number of output, input ports and qubits

Partition strategy/ceiling strategy

46% avg reduction across benchmarks (various quantum algorithms)

TopoLS Can compile circuits into tqec format

Future work: finding correlation surfaces

Q: Can you relax the grid of logical qubits to reduce the input space?